

0 3 . 1 Draw **and** label a single DNA nucleotide.

[2 marks]

0 3 . 2 Give **two** features of DNA **and** explain how each one is important in the semi-conservative replication of DNA.

[2 marks]

1 _____

2 _____



0 3 . 3

Replication of mitochondrial DNA (mtDNA) is different from that of nuclear DNA.

The replication of the second strand of mtDNA **only** starts after two-thirds of the first strand of mtDNA has been copied.

A piece of mtDNA is 16 500 base pairs long and is replicated at a rate of 50 nucleotides per second.

Tick (✓) the box that shows how long it would take to copy this mtDNA.

[1 mark]

- A** 330 seconds ☐
- B** 440 seconds ☐
- C** 550 seconds ☐
- D** 660 seconds ☐

5

Turn over for the next question

Turn over ►

